

YANYAN ALEXANDRA MO

Computational Research Associate
Stanford University

Dissecting gene regulatory mechanisms through functional genomics approaches

✉ ymo@stanford.edu  [Personal website](#)  [Github](#)

EDUCATION

M.S. in Biomedical Engineering

Aug. 2023 - May 2025

Johns Hopkins University

Baltimore, MD

- Dean's Master's Scholarship; Biomedical Engineering Departmental Award

B.S. in Computer Science | Chemical & Biomolecular Engineering

Aug. 2020 - May 2023

Johns Hopkins University

Baltimore, MD

- Graduated with Honors in both degrees

SKILLS AND INTERESTS

Research interests: Gene regulatory network inference, CRISPR-based perturbation screens (Perturb-seq), gene program discovery, agentic AI for biological interpretation

Programming languages: Python, R, C, C++, Java, Bash, HTML

Software engineering: Software and agentic system development, GPU computing, Git, AWS/HPC clusters

Other computer skills: L^AT_EX, Adobe Illustrator, Microsoft Office

RESEARCH EXPERIENCE

Computational Research Associate

July 2025 - Present

Department of Genetics, Stanford University

Stanford, CA

Advisor: J.Engreitz

- **PerturbNMF:** developed an end-to-end pipeline porting perturb-seq datasets to GPU-accelerated consensus non-negative matrix factorization (cNMF) to infer, evaluate, and interpret gene programs automatically.
- **Agentic AI systems:** built an LLM-based agentic framework for automated gene program annotation and literature-grounded mechanistic hypothesis generation
- **Biological discovery:** applied PerturbNMF to a Perturb-seq dataset of hiPSC-to-endothelial cell differentiation, identifying a novel role for Hippo signaling and charting its effects on cell fate across differentiation time
- **Consortium analysis:** deployed the PerturbNMF package as core components for IGVF and MorPhiC consortium datasets; presented results at both consortium meetings

Graduate Research Assistant

Oct. 2023 - June 2025

Department of Biomedical Engineering, Johns Hopkins University

Baltimore, MD

Advisor: P.Cahan

- **CRE-Finder:** developed an information-theoretic algorithm for reconstructing dynamic gene regulatory networks and predicting distal cis-regulatory elements from single-nucleus multi-omics data

Graduate Research Assistant

Mar. 2024 - June 2025

Department of Biomedical Engineering, Johns Hopkins University

Baltimore, MD

Advisor: W.Grayson

- **ARPA-H proposal:** conceived and drafted on predicting craniofacial injury healing models from time-series imaging, 3D microCT, and clinical records; conducted feasibility tests with machine learning models predicting bone and vascular healing patterns.

Undergraduate Research Assistant

Jan. 2023 - May 2023

Department of Computer Science, Johns Hopkins University

Baltimore, MD

Advisor: C.-M.Huang

- **Modeling human reliance on AI:** applied statistical testing and multi-step linear regression for feature selection and modeling to predict human reliance on AI from time-series eye-gaze data; second author, peer-reviewed conference paper

PUBLICATIONS

- **Mo, Y. A.**, Pushkarev, O., Fan, I., Gupta, R., Nguyen, T., King, H., Zeng, T., Everitt, A., Kang, H., Klie, A., Carter, H., Loh, K. M., & Engreitz, J. M., PerturbNMF: an end-to-end pipeline for scalable gene program analysis. *In preparation* (2026) [\[GitHub\]](#)
- Ma, X. R.^{*}, Conley, S. D.^{*}, Chen, M. S.^{*}, ... **Mo, Y. A.**, ... Goodyer, W. R.[†], & Engreitz, J. M.[†], Interpreting variants associated with congenital heart disease with a regulatory map of the human fetal heart. *Submitted to Nature* (2026)
- Cao, S., Ke, S.^{*}, **Mo, Y. A.**^{*}, Huang, C., & Liu, A., Eyes are the windows to AI reliance: toward real-time human-AI reliance assessment. *CHI Conference on Human Factors in Computing Systems* (2023) [\[PDF\]](#)

GRANTS & FUNDING

UCSF QBI Grant

Sept 2026

\$10,000 — Lead Applicant

San Francisco, CA, USA

- Awarded to develop AGenTic, an agentic LLM framework for automated gene set analysis, from a second-place hackathon prototype into a peer-reviewed publication

INVITED PRESENTATIONS

IGVF Consortium Annual Meeting

Sept 2026

Impact of Genomic Variation on Function

St. Louis, MO, USA

- Oral presentation: “Insights into Genome Function — Molecular Networks”

MorPhiC Consortium Face-to-Face Meeting

April 2026

Molecular Phenotypes of Null Alleles in Cells

Chicago, IL, USA

- Oral presentation: “PerturbNMF: An End-to-End Pipeline for Scalable Gene Program Analysis”

POSTER PRESENTATIONS

IGVF Consortium Annual Meeting

Sept 2025

Impact of Genomic Variation on Function

Durham, NC, USA

- Poster: “Performance evaluation of the GPU-accelerated consensus non-negative matrix factorization”

BMES 2024

Oct 2024

Biomedical Engineering Society Annual Meeting

Baltimore, MD, USA

- Poster: “Discovery of candidate cis-regulatory elements from single-nucleus multi-omics data with CRE-Finder”

AWARDS, SCHOLARSHIPS, AND HONORS

Stanford CVI Travel Award

Aug. 2026

Stanford Cardiovascular Institute

Stanford, CA, USA

- \$750 to present PerturbNMF at the IGVF Consortium Annual Meeting, St. Louis, MO

Second Place, QBI Hackathon

June 2026

UCSF Quantitative Biosciences Institute

San Francisco, CA, USA

- \$2,000 for AGenTic, an agentic LLM framework for automated gene set analysis

Dean’s Master’s Fellowship

Sep. 2023 - June 2025

Whiting School of Engineering, Johns Hopkins University

Baltimore, MD, USA

- \$32,000 per year for excellence in academic performance

Biomedical Engineering Departmental Award

Sep. 2024

Department of Biomedical Engineering, Johns Hopkins University

Baltimore, MD, USA

- Awarded for excellence in research

Best Project Award

May 2023

Intuitive Surgical Industry Showcase, Johns Hopkins University

Baltimore, MD, USA

- \$200 for the poster “Fine-tuning BERT for measuring linguistic uncertainty”

Center for Student Success Grant

Mar. 2022

Life Design Lab, Johns Hopkins University

Baltimore, MD, USA

- \$1,000 supporting a 2022 summer research experience

TEACHING EXPERIENCE

Computational Teaching Assistant

Sept 2023 – May 2025

Artificial Intelligence; Intermediate Programming

Computer Science, Johns Hopkins University, Baltimore, MD